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Can simple exchange heuristics guide us in predicting magnetic properties of solids?

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Keywords

Heuristics, Machine Learning, Magnetic Properties, Materials Informatics, Solid-state structures

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Can simple exchange heuristics guide us in predicting magnetic properties of solids?

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Abstract

The Kanamori-Goodenough-Anderson rules are a textbook heuristic for predicting magnetism. They connect bond angles to magnetic ordering for some transition metal compounds. Such domain knowledge is of high importance for building predictive machine learning models in scenarios with scarce data. Yet, there has been no statistical, large-scale evaluation of the heuristic. Here, we evaluate this heuristic on an experimental database of magnetic structures. We observe that the heuristic is largely satisfied, and we discuss the exceptions. We then demonstrate how integrating this heuristic into machine learning models for predicting magnetic ordering enhances prediction quality. Notably, these magnetism models are also capable of predicting if non-collinear magnetic ordering might occur. Furthermore, the heuristic provides a useful benchmark for evaluating theoretical methods that calculate magnetic properties.

Introduction

Developing new magnetic materials is crucial for transitioning to more sustainable technologies. The permanent magnets currently in use rely heavily on scarce and environmentally problematic rare-earth (RE) elements. Identifying promising alternative materials from a seemingly endless number of possibilities has been increasingly addressed by data-driven approaches, and many machine-learning (ML) studies have been concerned with predicting the magnetism of crystalline solids.[1–7]

Within the ML of materials properties, finding a sensible materials representation is essential. Integrating domain knowledge, such as heuristics related to structure-property relationships, into developing materials descriptors (features) presents a natural approach.[8–10] This technique has been successfully applied in various contexts, such as predicting perovskite stability from the ABX_3 composition. By following a famous heuristic for perovskites, the Goldschmidt tolerance factor,[11] many ML studies successfully employed features derived from ionic radii in this classification task and improved on the accuracy of the original heuristic.[12–16]

In the context of RE-free magnetic materials, a well-known structure–property heuristic are the Kanamori-Goodenough-Anderson (KGA) rules (Fig. 1 a).[17–20] The dominating exchange interaction in ionic and insulating transition-metal (TM) compounds usually is superexchange, an exchange interaction between magnetic sites mediated by connecting ligand orbitals. The KGA rules predict the sign (antiferromagnetic (AFM) or ferromagnetic (FM)) and strength of superexchange interactions. They are stated below:

1. The exchange interaction between two half-filled d orbitals with non-zero overlap via one ligand (L) p orbital is strong and AFM.

2. The exchange interaction between a half-filled and an empty or a half-filled and a double-filled d orbital with non-zero overlap via one ligand (L) p orbital can be both AFM and FM, but is usually weakly FM due to an additionally stabilizing intra-atomic exchange term.

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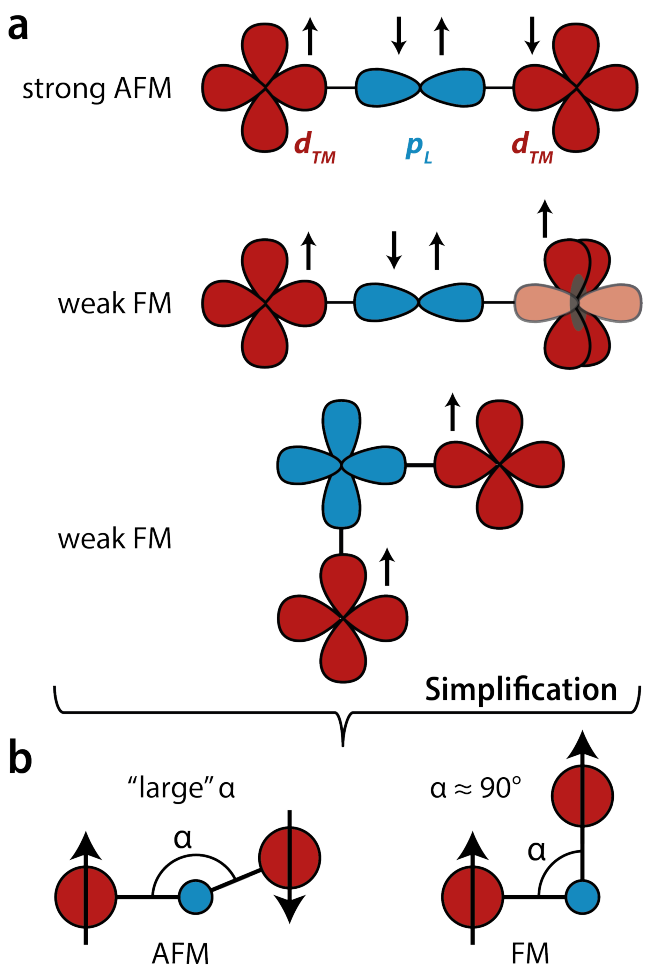

Figure 1: Schematic display of the KGA rules of superexchange (a) and their simplification considering bond-angle information only (b). TM d orbitals and atoms are depicted in red, ligand p orbitals and atoms are depicted in blue.

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3. Exchange interactions between two half-filled d orbitals that involve two L p orbitals are weakly FM.

Although no large-scale evaluation of these rules exists so far, they are widely regarded as well-satisfied and are a regular part of textbooks on magnetism.[21,22] Some sources derive further simplifications of the KGA rules[22–24] for common cases, such as nearest-neighbor interactions of octahedral TM sites that are the same ion. In these cases, mostly KGA rules (1) and (3) apply, so these simplified rules of thumb only consider the bond angle between neighboring magnetic sites (Fig. 1 b): Larger TM–L–TM bond angles tend to favor AFM spin orderings, 90 ° TM–L–TM bond angles are typically associated with FM orderings.

There are multiple exceptions to these simplifications: By neglecting orbital information, the rules of thumb do not account for the cooperative Jahn-Teller effect and orbital ordering. In the case of antiferrodistortive orbital ordering, an often-observed FM ordering of Jahn-Teller-active ions is expected as per KGA rule (2). The rule of thumb predicting FM exchange at 90 ° bond angles is oversimplifying as well. Although based on KGA rule (3), many different exchange interactions can possibly occur at 90 ° TM–L–TM bond angles: Apart from FM superexchange via two L p orbitals, superexchange may occur via only one L p orbital, resulting in either AFM or FM ordering.[21] In addition, d–d overlap and direct exchange may be significant at 90 ° bond angles.[21,23,25–27]

Nevertheless, the rules of thumb remain valid for common cases: E.g., in the common case of 180 ° spin orderings between identical and octahedral ions, the rules of thumb predict AFM exchange, while the original KGA rules almost always do as well. Based on both the original and simplified KGA rules, TM–L–TM bond-angle-derived descriptors, potentially together with other elemental features, could serve as a valuable feature in ML studies of magnetic structures, as they would contain implicit physical knowledge while being computationally cheap, i.e., no additional quantum-chemical calculations are required.

However, a previous ML study of ionic TM compounds from DFT + U calculations of the *MaterialsProject*[28,29] (MP) found no importance of TM–L–TM bond-angle-derived features for predicting the magnetic structure.[2] This could be because either the KGA rules or the KGA rules of thumb are too strong a simplification. Alternatively, it could be because the level of theory of the high-throughput DFT + U methodology[2,30] and the magnetic structural models (missing non-collinearity) are insufficient in capturing magnetism–bond angle characteristics present in many experimentally observed magnetic structures.

To investigate whether the KGA rules can guide us in predicting magnetism, we analyze nearest-neighbor spin ordering trends in the crowd-sourced MAGNDATA database,[31,32] the largest collection of experimentally determined magnetic structures. We evaluate bond angle distributions between magnetic sites as a function of the angle between magnetic moment vectors. We want to determine whether the KGA rules of thumb are well-fulfilled for this dataset and identify possible reasons for any violations. We repeat the same analysis for the above-mentioned MP dataset of DFT + U-computed magnetic structures[2,30] and compare the results. We also revisit aspects of the original KGA rules through an ion pair-resolved analysis.

In the second part of our study, we build an ML model for magnetic structures based on compositional and structural descriptors. We introduce a new, scalar magnetism label that includes non-collinear structures and accounts for certain peculiarities of experimental magnetic structures. We additionally provide new ML models for magnetic structures that can predict certain non-collinear magnetic structures.

Results and discussion

Experimental data pre-processing

MAGNDATA contains many groups of magnetic structures that share the same parent structure, which is defined solely by atomic positions and unit cell parameters, excluding magnetic moment vectors. This can be due to the existence of different magnetic phases at different temperatures or even due to multiple competing magnetic models for the same magnetic phase. As materials descriptors for machine learning are derived from the non-magnetic parent structure, our dataset would contain (feature − target) datapoints with the same features, leading to falsely worsened or improved model predictions. To prevent this, we filter the commensurate magnetic structures from MAGNDATA to retain a subset of magnetic structures with unique parent structures. We further filter for entries that are ionic enough to allow for automated assignment of cations and anions, and determine their connectivities using *ChemEnv*[33] (see SI Sec. 2.1.1 for methodology details). This leaves 938 entries. Bond angle occurrences between magnetic sites are analyzed as a function

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of the angle between the magnetic moment vectors of neighboring magnetic sites. Although non-collinear nearest-neighbor spin orderings are present in the dataset, especially in those containing RE sites, we focus on collinear orderings in our analysis.

Statistical analysis

Fig. 2 shows the bond angle distributions of FM and AFM spin orderings of octahedral TM sites of the same ion in MAGNDATA. This is the subdataset to which the KGA rules of thumb apply. We note that very similar trends are observed upon changing the analysis parameters, such as normalizing bond angle occurrences to 1.0 per structure or only considering oxygen-connected TM octahedra (see SI Fig. 2). FM spin orderings occur predominantly around 90 ° bond angles, with 66 % of all FM bond angles in (90 ± 15) °. This trend aligns well with the KGA rule of thumb, which states that FM interactions typically occur at 90 ° bond angles.
79 % of all collinear spin orderings with bond angles greater than 110 ° are AFM, supporting the KGA rule of thumb that larger bond angles correspond to AFM interactions. However, 35 % of all AFM orderings center around 90 ° bond angles, which the KGA rules of thumb do not predict. A detailed discussion on this topic is included in the SI Sec. 2.1.3. Notable structural differences between AFM and FM orderings around 90 ° are observed, especially in the TM–TM distance and the degree of distortion in the coordination octahedra. Inferred purely from structural data, likely reasons for rule violations include AFM direct exchange, the breakdown of the rules due to deviations from perfect symmetry, and competing superexchange pathways. The fact that AFM spin orderings can occur at 90 ° bond angles likely reduces the suitability of bond angles as features in ML models. However, if the rule violations can be inferred from structural effects, an ML model that also has access to this structural data might still learn to discriminate between competing magnetic interactions at 90 ° bond angles. Regardless of the underlying exchange interactions, the overall bond angle trends in the FM and AFM spin orderings depicted in Fig. 2 are different.

We also observe that FM and AFM orderings show very different bond angle patterns when they occur between connected TM atoms in non-octahedral environments. Interestingly, bond angle trends observed for collinear spin orderings of octahedral RE sites of the same ion differ significantly from the trends shown in Fig. 2 (see SI Fig. 8), hinting at different underlying exchange heuristics. However, only 37 of 938 structures contain this motif, significantly limiting the generalizability of this observation.

Further, we perform a bond angle analysis on the DFT + U dataset of the MP from 2020 that was used in the ML study that found bond-angle derived features to be irrelevant.[2] They applied a Hubbard U correction to the PBE exchange functional[34] to account for the strong Coulombic interaction between localized electrons of several $3d$ elements, using a set of standard U values determined by minimizing the error in a set of known binary formation enthalpies.[35–37] The dataset was obtained by enumerating several possible collinear magnetic configurations per structure, defining the configuration with the lowest DFT energy as the magnetic ground state.[30]

Contrary to the MAGNDATA database, the bond angle distributions of the FM and AFM spin orderings between TM octahedra of the same ion in the MP dataset are highly similar (see SI Fig. 9). 62 % of all FM bond angles and 59 % of all AFM bond angles center around (90 ± 15) °. The fraction of bond angles greater than 110 ° is also very similar, with 37 % in the FM spin orderings and 39 % in the AFM orderings.

In SI Sec. 2.3, we further compare the bond angle distributions in the experimental and theoretical datasets using the Kolmogorov-Smirnov test.[38,39] Test statistics and p values are discussed as a function of the number of structures to account for the different dataset sizes.[40,41] Test statistics are $\sim 3 - 6$ times larger for experimental structures than for DFT + U structures, which agrees with our visual observations that FM and AFM bond angle distributions are more dissimilar for experimental magnetic structures.

Since the publication of the DFT + U database in 2020, the MP has calculated the magnetic structures of further materials with the same methodology. We also note that the statistical test results for all magnetic structures that are available via the MP API are very similar to those structures published in the 2020 database[2] (see SI Sec. 2.3).

A direct comparison of MP and MAGNDATA is hindered by dataset differences, as detailed in SI Sec. 2.3. The MP dataset contains a higher fraction of mixed-valence compounds and ions well-known for orbital ordering, which means that the KGA rules may not always be applicable or could appear unviolated in some instances. Nevertheless, analysis of ions without orbital ordering and comparison of the common subset of

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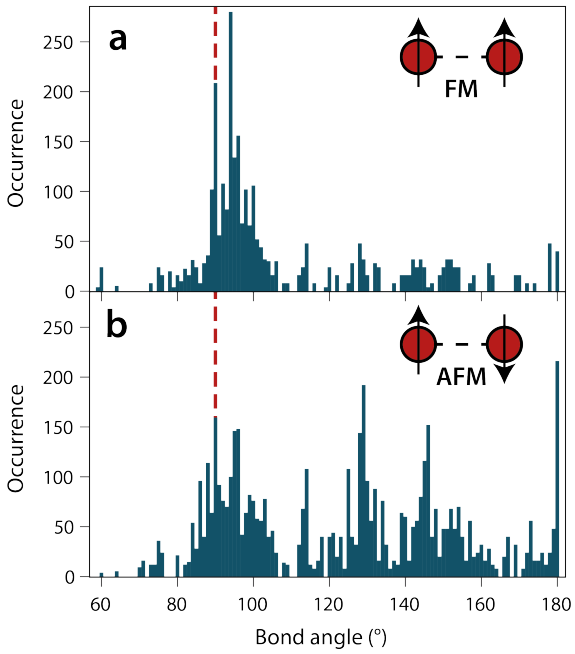

Figure 2: Bond angle distributions of FM (a) and AFM (b) spin orderings of octahedral TM sites that are the same ion in the structurally unique, commensurate MAGNDATA subset. The red, dashed line marks 90° bond angles. Absolute bond angle occurrences within the magnetic primitive cell are shown. 336 structures contribute to this figure.

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both datasets suggest that the applied DFT + U-based methodology is insufficient to capture the exchange characteristics of many real-world magnetic structures.

A previous methodology benchmark on 64 commensurate and collinear MAGNDATA entries resulted in 61 % of the cases in the correct magnetic structure.[30] Accurately predicting the magnetic ground state of any compound would require multiple modifications to the current workflow. Beyond incorporating non-collinear magnetic structures into the enumeration, the inclusion of screening interactions between spin channels via Hund J values was shown to be crucial for modeling non-collinear magnetic structures.[42,43]

Additionally, Hubbard U parameters vary strongly per element among large datasets when computed from first-principles,[43,44] and energy differences between magnetic configurations were demonstrated to depend on U and J parameters.[42,43,45] Even when initialized with experimental magnetic moments, DFT+U calculations were shown to converge to different magnetic ground states as a function of U.[46] Using fixed U values fitted to binary formation enthalpies very likely introduces errors, especially as multiple magnetic orderings were reported to have very similar DFT energies in the aforementioned benchmark of 64 structures.[30]

Further, including spin-orbit coupling would be especially relevant for magnetic systems containing sites with larger atomic numbers and unquenched orbital moments. Also, recent developments in source-free exchange-correlation functionals[47–49] improved the description of magnetic moments and non-collinearity in various bulk systems.

However, we also note several limitations of crowd-sourced experimental databases like MAGNDATA: The resolution and quality of entries vary, e.g., due to differences in the experimental setup or the experimentalist's experience. Not all possible magnetic models may have been taken into account in the determination of the magnetic structure, as noted in the MAGNDATA database for several entries. Additionally, the atomic positions of some entries originate from measurements above the magnetic phase transition temperature. As magnetostructural coupling is usually very weak,,[31] this might not present a significant problem. Still, general temperature effects might influence our bond-angle analysis. Most importantly, MAGNDATA naturally has the huge bias of containing only phases experimentalists considered promising and interesting enough to investigate with neutron sources, and that have magnetic structures sufficiently interesting to be published.

Coverage of larger chemical spaces with constant scanning conditions, a minimized "bias of interestingness" and realistic results can likely be achieved best by combined computational and experimental high-throughput studies. Our analysis emphasizes the need for high-quality computational magnetism data that captures essential exchange characteristics present in real-world magnetic structures. Additionally, such computational data would allow for a full review of the original KGA rules, which rely on orbital symmetry and occupancy arguments.

Before we turn to the ML section, we briefly revisit the original KGA rules using the limited data available. Although orbital occupation and symmetry are inaccessible, we can roughly estimate electronic configurations of octahedral TM sites from their automatically determined oxidation states. Fig. 3 a displays the ion pair-resolved ratio of FM / AFM spin orderings of octahedral TM sites at large bond angles ($> 150\,^\circ$). Many observations align with the KGA rules: for instance, d^3–d^3, (presumably) high spin-d^5–d^5 or d^8–d^8 pairs predominantly exhibit AFM orderings.[19,53] Rocksalt-type CrN presents an interesting exception (Fig. 3 c). Its $180\,^\circ$ Cr^{3+}–N–Cr^{3+} FM spin orderings are attributed to partially covalent Cr–N bonding.[51,53]

Many off-diagonal elements depicted in Fig. 3 are ion pairs with and without e_g electrons, for which the second KGA rule predicts FM coupling. Deviations observed in our dataset are well-known exceptions to the KGA rules: $3d5d$ compounds with poor $3e_g$–$5e_g$ orbital overlap. Magnetism in these systems is often explained by structural distortions that enable otherwise symmetry-forbidden superexchange pathways, e.g., $3e_g$–$5t_{2g}$. One such case is Ca_2NiOsO_6 (Fig. 3 b).[50] Further analysis of bond angle–magnetism trends of ions with and without e_g electrons is provided in SI Chap. 2.4.

Ions susceptible to orbital ordering present further interesting cases. The pair Mn^{3+}–Mn^{3+} is present in 9 structures and displays roughly 3 times more FM than AFM spin orderings at large bond angles in the dataset. Consistent with the KGA rules, nearest-neighbor FM spin orderings are generally observed with antiferrodistortive orbital ordering, and vice versa. The types of orbital ordering vary across the structures, such as the 4 antiparallel and 2 parallel aligned neighbors per elongated octahedra in entry #0.1, $LaMnO_3$,[54] or the exclusively parallel alignment of elongated octahedra in entry #0.823, brownmillerite-type Sr_2MnGaO_5.[55] Since the orbital ordering is assumed but not derived from the KGA rules, predicting magnetism in theoretical compounds requires additional heuristics.

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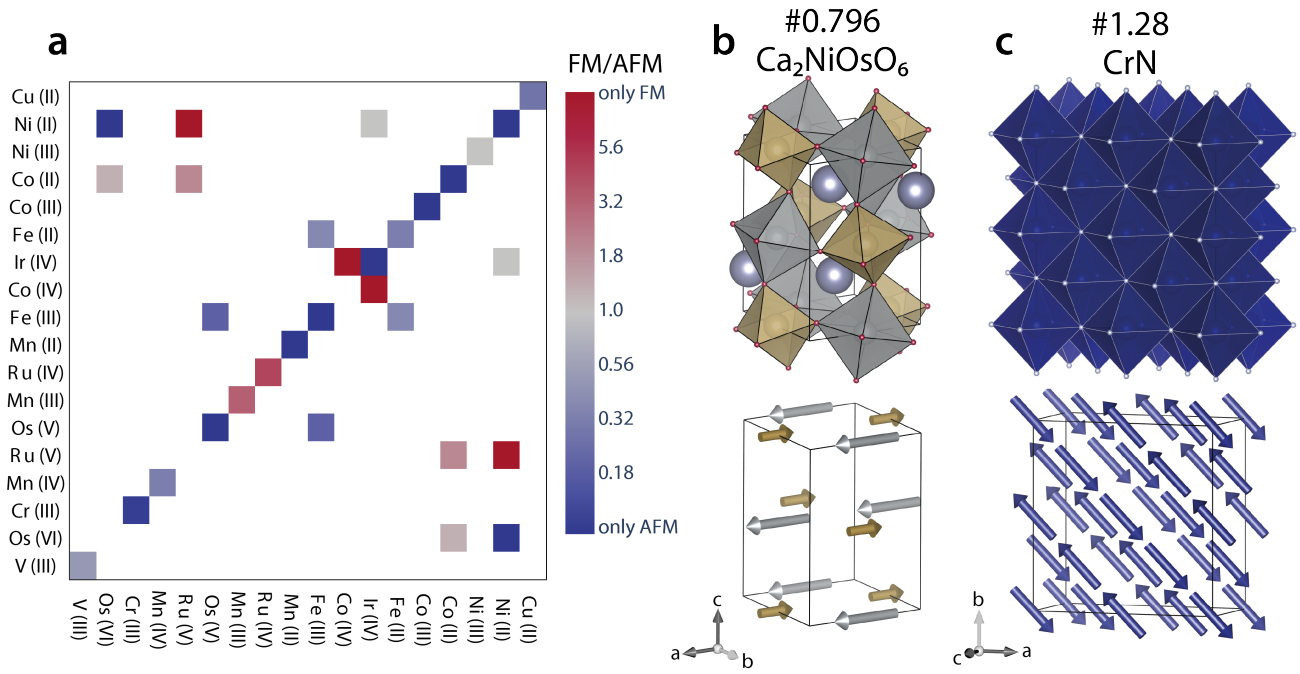

Figure 3: (a) Ion-pair-resolved, logarithmic heatmap plot of the ratio of FM / AFM spin orderings with bond angles greater than 150° of the octahedral TM sites in the crystallographically unique, commensurate MAGNDATA dataset. Absolute occurrences per magnetic primitive cell are displayed, and ions are arranged along the axes based on their estimated electronic configurations. 92 structures contribute to this figure, observed trends are similar upon only including oxygen-mediated spin orderings. SI Figs. 12 and 13 depict the absolute number of FM and AFM orderings per ion pair and the number of structures contributing to each datapoint. (b) MAGNDATA entry #0.796, double perovskite Ca_2NiOsO_6,[50] that contains ≈ 150° AFM Ni^{2+}–O–Os^{6+} orderings. Crystal structure depiction similar to the original study.[50] (c) MAGNDATA entry #1.28, rocksalt-type CrN,[51] that contains 180° Cr^{3+}–N–Cr^{3+} FM orderings. Crystal structures visualized with VESTA.[52]

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Only 35 % of ion pairs in Fig. 3 a exhibit FM / AFM ratios between 0.32 and 3.2 (i.e., comparable FM and AFM ordering occurrences) for large bond angles. The participating ions are a good predictor of the sign of the spin ordering of octahedral TM sites in this case. This contrasts with the 90 ° case (SI Fig. 14), where spin orderings are less dependent on the interacting species. This matches the higher prediction uncertainty for 90 ° orderings reported by Kanamori and Goodenough.[19,53]

Our approach has several limitations. Automatically assigned oxidation states might in some cases not match experimental observations, as in entry #0.747, $Ba_3CoIr_2O_7$, where our bond valence sum analysis indicates Ir^{4+} and $Co_4{}^{+}$, while XPS measurements yield Ir^{5+} and Co^{2+}.[56] Additionally, Fig. 3 contains datapoints where our naive assumption of insulators is incorrect, such as the metal $SrRuO_3$ (entry #0.732).[57] Most importantly, the low number of 1–14 structures per ion pair (see SI Fig. 13) limits the generalizability of our findings.

Further aspects of the original KGA rules are discussed in SI Chap. 2.4, including analyses of spin orderings at 90 ° bond angles, e_g–non-e_g spin orderings, and two case studies on bond angle distributions for well-studied ions: Cr^{3+} ($t_{2g}^3 e_g^0$) and Fe^{3+} / Mn^{2+} ($t_{2g}^3 e_g^2$).

Finally, we briefly note that the characterization and prediction of the 2D magnetic structures is a very active field of research. The applicability of the KGA rules for 2D materials is referenced in many studies.[23,27,58–60]

Machine-learning model for magnetism

As an extension of our statistical analysis above, we train ML models on magnetic structures from experiments and calculations. We analyze model performances and feature importances to answer the title question of this paper - whether the KGA rules of thumb can guide us in predicting magnetic structures.

The full magnetic structure, i.e., all atomic magnetic moment vectors in the magnetic unit cell, is a very complex label (target) and unsuitable for ML studies with sparse training data. Instead, magnetic structures are often learned by classification of the two labels AFM and FM.[2,5–7] However, this approach is only suited for collinear magnetic structures. A naive alternative to label partially non-collinear datasets like the MAGNDATA database would be to add a third, non-collinear class. Still, small, non-collinear contributions to the magnetic structure (e.g., an additional perpendicular FM component due to the Dzyaloshinsky-Moriya interaction[61,62] (DMI)) might not be resolved in the neutron diffraction experiment. These contributions might go unnoticed or only be detected through additional magnetization measurements and may not be explicitly recorded in the database entry. In many cases, the labeling would be a function of the experimental setup. Total magnetization is also not a suitable target, as often only the directions of atomic magnetic moments in MAGNDATA are known from experiment, and the magnitudes may be set to arbitrary values. Other possible targets are the magnetic propagation vector(s) k, which describe the lowering of translational symmetry due to magnetic ordering and can classify arbitrarily complex magnetic structures. A zero/non-zero classification model for propagation vectors was trained in a previous study.[5] However, propagation vectors alone do not contain information on many application-relevant aspects of the magnetic structure.

Given this situation, we introduce a new magnetism label that is low-dimensional, high in information, and can characterize all our datasets: the parallelity score p. We define the structure-wise p score as the fraction of all nearest-neighbor spin orderings of magnetically ordered sites that are FM orderings,

$$p_{struct} = \begin{cases} \frac{n(mi_{FM})}{n(mi)} & \text{if } n(mi) > 0 \\ -1 & \text{otherwise,} \end{cases} \tag{1}$$

with $n(mi)$ being the number of all magnetic nearest-neighbor spin orderings in the structure and $n(mi_{FM})$ the number of FM orderings.

Fig. 4 depicts the determination of the p score (not to be confused with the p value of the statistical analysis mentioned above) for a hypothetical 2D magnetic structure. There are three nearest-neighbor magnetic spin orderings, out of which two (named (1) and (3)) are FM spin orderings. Therefore, the structure-wise p score is 2/3.

An antiparallelity score ap can be defined accordingly,

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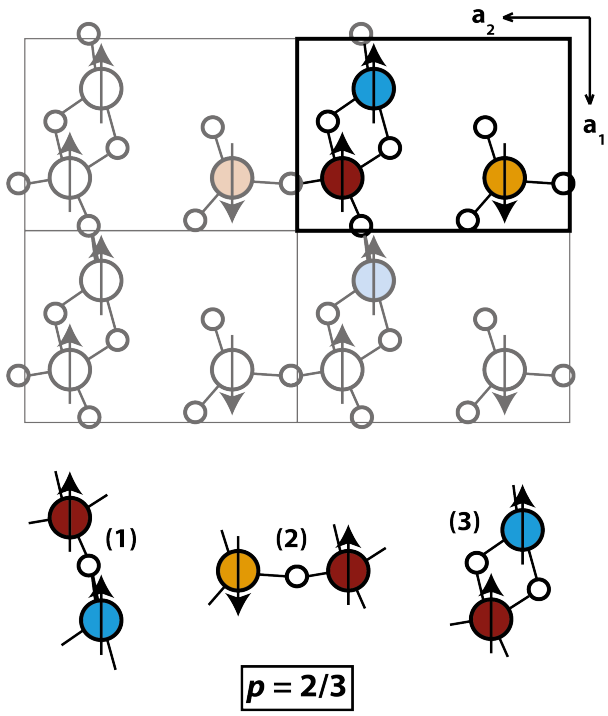

Figure 4: Schematic representation of the p score determination for a magnetic structure. Magnetic sites and their ligand atoms are depicted in the magnetic unit cell and neighboring cells (in gray).

$$ap_{struct} = \begin{cases} \frac{n(mi_{AFM})}{n(mi)} & \text{if } n(mi) > 0 \\ -1 & \text{otherwise.} \end{cases} \tag{2}$$

The structure-wise ap score in Fig. 4 is $1/3$.

As connectivity in crystalline solids can be ambiguous, our proposed labeling is only unique for a specific neighbor determination method. Here, we use the same *ChemEnv*[33]-based approach as in the statistical analysis, further described in SI Sec. 2.1.1. We also demonstrate in SI Sec. 3.2 that the labels in our dataset are almost constant upon changing the parameters for determining neighboring atoms.

The summation over all magnetic nearest-neighbor spin orderings should present a non-local view of magnetic structures. In addition, even itinerant magnetism can be predicted from the local bonding situation.[63] Other neighbor determination methods for label computation, e.g., based solely on distance arguments, are also possible.

We note that $n(mi)$ can be zero with our neighbor determination approach. This corresponds to structures with magnetic sites that do not share common ligands with other magnetic sites and where other exchange mechanisms might be present, such as super-super exchange via multiple ligands. We define these structures' p and ap as -1, and focus on structures with a p or ap score ≥ 0 in this work. The limitations of the current p score implementation are discussed in SI Sec. 3.2.

To eliminate small non-collinear contributions to the label, which can be a function of experimental setup, we include the possibility of adding an angle tolerance to count two magnetic vectors as aligned parallel in the determination of p. We use an angle tolerance of $10\,^{\circ}$ in this work unless otherwise stated.

A single magnetic site can be labeled as well by filtering Eq. 1 for all spin orderings that include this specific site:

$$p_{site} = \begin{cases} \frac{n(mi_{site,FM})}{n(mi_{site})}, & \text{if } n(mi_{site}) > 0 \\ -1, & \text{otherwise} \end{cases} \tag{3}$$

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with n(mi_{site}) as the number of all nearest-neighbor spin orderings that include the site.

From Eqs. 1 to 3 it follows that $p + ap = 1.0$ only holds true for collinear structures and sites. The sum of both scores is not only suited for the prediction of (highly) non-collinear structures but also provides an entry point for the high-throughput analysis of "non-collinear sites" (i.e., sites with neighbors with magnetic vectors that are (partially) not aligned collinearly) and non-collinear structures. An outlook on the systematic analysis of these compounds is given at the end of this paper. We also note that the site-wise p score provides an entry point for the machine learning of local phenomena such as magnetism at defects or grain boundaries.

For the dataset of 938 ionic, unique, commensurate MAGNDATA structures, 802 magnetic structures have a structure-wise p and ap score ≥ 0, i.e., contain magnetic sites that share ligand atoms with neighboring magnetic sites. A subset of 447 structures is RE-free. As this subset is chemically more similar to the MP database (and more closely related to the KGA rules of thumb), we train models on both this subset and the set of all 802 MAGNDATA structures. In the MP database, our labeling approach gives 2013 structures with a structure-wise p score ≥ 0. The label distributions of all three datasets are shown in SI Figs. 17–19. Considering the trimodal label distributions and the small dataset sizes, we bin the distribution into three classes: $0 \leq x \leq 0.005$, $0.005 < x \leq 0.995$, and $0.995 < x \leq 1.0$ with x in p, ap, resulting in a multi-class classification problem. For the collinear MP dataset, the sum of the structure-wise p and ap scores is always 1.0, and a complete mapping between the binned p and ap labels exists. Therefore, we only learn this target once and include it in further discussions on both p and ap targets.

We engineer magnetism-specific structural features, such as bond angles, electronegativities, or distances between sites that are guessed to order magnetically. The guessing has to be performed as the information about which sites would order magnetically is not known prior to the experiment, and featurizing the true magnetic sites would be a form of data leakage. SI Sec. 3.3 gives an overview of our features and guessing methods.

Not only descriptors derived from bond angles, but several other magnetism-specific features are related to the KGA rules of thumb. One example is the coordination polyhedra connectivity: Further analysis of the spin orderings shown in Fig. 2 reveals that 78 % of all corner connections between collinearly ordered, octahedral TM sites of the same ion are AFM. In principle, the KGA rules of thumb could be reformulated as a function of connectivity instead of bond angle. Different possible TM–TM exchange interactions as a function of connectivity have also been summarized elsewhere.[64] Additionally, bond angles between magnetic sites are highly correlated with their interatomic distance. This already points out an issue in answering our title question: The distance between magnetic sites can encode much information related to the KGA rules of thumb. However, the distance also comprises information on other exchange interactions, such as direct exchange. For now, we declare all features derived from bond angles, connectivities, and distances of sites guessed to order magnetically as KGA-related.

In addition to our magnetism-specific descriptors, compositional and structural Matminer[65] features are applied. Before further analyses and model training, we split each dataset 20 times into 95 % train and 5 % test sets with no overlap between the test sets. This yields 20 models per dataset and target. To exclude dataset size effects from the model comparison, we subsample all train sets to be of the same size as the train sets of the smallest dataset, the 447 RE-free structures of MAGNDATA.

The statistical analysis above analyzes single bond angles between octahedral TM sites of the same ion. However, both our target and our KGA-related features in the ML approach are structure-wise statistics that consider the whole magnetic structure. Additionally, as magnetic sites are guessed, featurization adds noise compared to the actual statistics of magnetically ordering neighbors in a structure (see SI Sec. 3.3). Therefore, it is unclear whether the statistical trends above directly translate into higher KGA-related feature importances for the MAGNDATA databases compared to the MP database.

As feature importances are difficult to compare between models trained on different datasets, we first compare all feature–target normalized mutual informations (NMIs)[66–68] between the different datasets. Mutual Information can be thought of as a powerful correlation measure that also captures non-linear and non-monotonic relationships between variables. SI Sec. 3.4 contains more information on this metric and its normalized variant, the NMI. The NMI of two sampled distributions cannot be calculated exactly, but is estimated and depends on a random seed. For each feature in each of the 20 train splits of the three datasets, we compute the NMI with the binned p and ap target many times. Fig. 5 contains box plots of the highest mean feature–target NMI of a KGA-related feature per target and per dataset. Each box plot comprises 20 data points, each point representing a train split.

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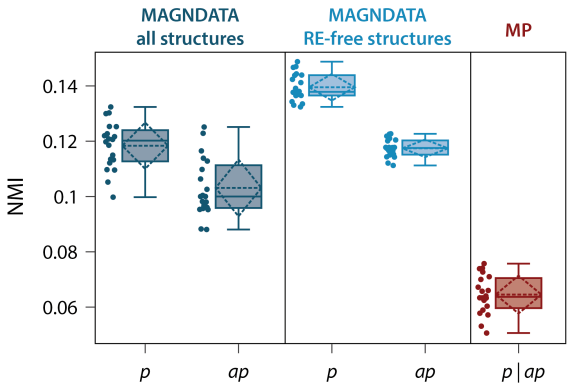

Figure 5: Box plots of the means of the highest feature–target normalized mutual information (NMI) of a KGA-related feature with the binned p and ap scores for the different datasets. Each box plot comprises 20 train splits. The dashed lines correspond to the mean and standard deviation of the distributions. NMI, a normalized measure for variable interdependence, is defined in detail in the SI Sec. 3.4.

Table 1: Average weighted test F1 scores of p and ap score models evaluated on the CV and hold-out test sets. Additionally, the difference from the respective baseline F1 is shown to account for class imbalance differences among the datasets and ensure comparability.

		CV test set		hold-out test set	
		F1	difference to baseline	F1	difference to baseline
MAGNDATA, all	p	0.69(2)	0.25	0.69(8)	0.25
	ap	0.64(3)	0.22	0.65(5)	0.23
MAGNDATA, RE-free	p	0.71(2)	0.27	0.70(7)	0.26
	ap	0.64(2)	0.23	0.62(7)	0.21
MP	$p\|ap$	0.61(2)	0.20	0.60(5)	0.19

In the magnetic structures from experiment, the feature–target NMIs are 1.6-2.2 times higher than in the MP database, indicating a stronger interdependence between the KGA-related features and the magnetism targets.

Box plots of the highest feature–target mean NMIs (not necessarily KGA-related features) display a similar trend as in Fig. 5 and are depicted in SI Fig. 23. It is noteworthy that in the case of the RE-free structures of MAGNDATA, the most important feature always corresponds to a KGA-related one. This is often, but not always, the case for the set of all structures of MAGNDATA and the MP dataset, with features related to atomic valences or oxidation states being the most important in some train splits.

In absolute terms, general magnetism-related and KGA-related features are more interdependent with the magnetism targets in the experimental datasets than in the DFT + U dataset, as is shown by significantly higher feature–target NMIs. Relatively speaking, KGA-related features are among the descriptors with the highest target interdependence in all three datasets.

For each dataset and target, 20 features are selected based on high feature–target NMIs and low feature–feature NMIs with the procedure implemented in the *modnet* package.[69] As the primary goal of our ML experiment is to compare model performances and feature importances among different datasets, we choose a comparatively simple yet interpretable Random Forest (RF) modeling approach. RF models are trained with a 10-fold nested cross-validation (CV) approach. Our approach involves ensembles of 20 RF models per target and dataset (see SI Sec. 3.4 for ML workflow details).

Fig. 6, Tab. 1, and SI Fig. 25 show model performances on the outer CV and hold-out test sets. The

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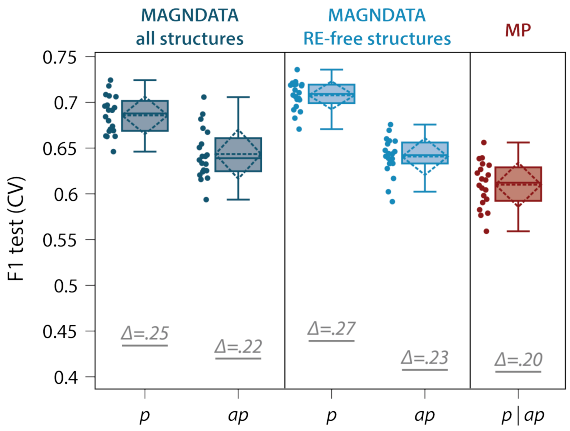

Figure 6: Box plots of the weighted F1 CV test scores for the p and ap multiclass models trained on the commensurate, structurally unique MAGNDATA dataset, its RE-free subset, and the unique MP dataset. The dashed lines correspond to the mean and standard deviation of the distributions. A direct p to ap target mapping exists for the collinear MP dataset. The baseline F1 and its difference from the mean F1 are given in grey. Each box plot comprises 20 models.

main evaluation metric is the weighted F1 score, which is used for multi-class classification problems and is especially suited for imbalanced problems. It can range between 0 and 1, with 1 denoting a perfect classifier. The difference from the respective baseline F1 is also given to account for class imbalance differences among the datasets. Both weighted F1 scores and differences to baseline F1s of models trained on experimental data are systematically larger than those of models trained on the MP dataset.

As dataset size effects are eliminated, we observe that including RE structures in the training on the MAGNDATA dataset has a slight adverse effect on the model prediction quality. Analysis of the TM-free MAGNDATA structures reveals high feature–target NMIs and high RF model prediction quality. Thus, even though many of the features were designed with TM-magnetism in mind, TM-free structures display high feature–target NMIs of general structural and elemental features and can be learned well by our models.

Among the most important features of all models are those expected to generally correlate with magnetism, e.g., descriptors derived from the number of valence electrons of the elements in the composition, as well as KGA-derived descriptors. This is consistent with the findings from the NMI analysis above. In all datasets, KGA-related features are among the descriptors that share the highest mutual information with the target, reflected in high KGA feature importances across all datasets. However, absolute feature–target NMIs are considerably lower in the MP dataset. This is reflected in the lower performances of models trained on the MP dataset as compared to models trained on experimental data.

To further answer our title question, we train comparison models without certain groups of features related to the KGA-heuristic, i.e., without descriptors derived from bond angles, connectivities, and distances of sites guessed to order magnetically. The overall number of features per model remains 20. We evaluate the worsening of the F1 test scores of the model ensembles with the one-sided Welch test[70] (see SI Sec. 3.6.1). Eliminating one of the three descriptor groups leads, with two exceptions, to insignificant decreases in model performance. This is because the shared information between these groups of features is relatively high, and they can replace each other. However, eliminating all three groups of features leads to significantly worse models. This is observed for all targets and databases' CV and hold-out F1 test scores, except for the hold-out F1 test scores of p score models trained on all MAGNDATA structures. When considering the CV test sets, the model worsening effect is by far the highest for the RE-free structures of MAGNDATA. The effect in the hold-out test sets is a bit larger in the MP database.

To summarize, feature–target NMIs, feature importances, and comparison models all demonstrate that descriptors derived from the KGA rules of thumb can guide us in predicting magnetic structures. From the

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statistical analysis above, this is expected for experimental magnetic structures. For the high-throughput DFT + U magnetic structures that did not show a fulfillment of the KGA rules of thumb in the statistical analysis above, this observation is a bit more surprising. However, lower model performances and lower feature–target NMIs of the DFT + U magnetism data hint at a weaker structure–property relationship. Additionally, the KGA rules of thumb reviewed above only describe the magnetism of octahedral TM sites of the same ion. It is possible that other bond-angle–magnetism trends might be present in the MP dataset, causing the significant bond-angle feature importances.

To shed further light on the differences from the work by Frey et al.,[2] we train an AFM / FM classifier on the MP dataset and compare it in detail to the study by Frey et al.[2] and our p and ap score models. Results, including feature–target NMIs, model performances, and feature importances, can be found in SI Secs. 3.5.2 and 3.6.2.

Conclusion

We use a dual approach combining statistical analysis and machine learning to assess the validity of a textbook magnetism heuristic that links bond angles and magnetism. The heuristic is evaluated on both experimental and theoretical DFT + U magnetic structures on the large scale for the first time. We observe that the heuristic is partially satisfied in magnetic structures from experiments, and we are able to rationalize the observed exceptions. However, we do not find the heuristic fulfilled in the theoretical DFT + U dataset. This study provides new insights into the applicability of magnetism heuristics for rational materials design, offering valuable guidance for chemists and materials scientists. We propose evaluating future computational magnetism workflows using the methodology presented in this work on structures that are chemically similar to MAGNDATA entries. Calculation results should display similar structure–property relationships as those of magnetic structures from experiment observed in this work. Establishing such benchmarks is essential for the development of theoretical methods that can reliably predict complex magnetic structures, a critical property in the design of functional materials.

Furthermore, we demonstrate that the heuristic can guide us in predicting magnetic structures: features derived from it show high importance, and including them leads to improved performance of the ML model. This provides valuable information for developing machine learning approaches to magnetism, and our results stand in contrast to previous studies relying on the DFT + U dataset. We accommodate certain peculiarities of magnetic structures from the experiment, mainly non-collinear magnetism, by introducing new, low-dimensional magnetism labels, the p and ap scores. Those labels are well-suited for data-scarce scenarios, likely for magnetism datasets.

Future work will focus on the combined prediction of multiple low-dimensional magnetism targets, such as structure- and site-wise p scores, and magnetic propagation vector(s) k, possibly achieving full magnetic structure prediction for certain structures.

Additionally, a combined p and ap score, representing a system's collinearity, could serve as a valuable target for future ML studies. As pointed out above, we define a magnetic site's or structure's collinearity c as the sum of p and ap, yielding a scalar value between 0 and 1. This compact definition not only makes c suitable as a machine-learning target but also facilitates the automated analysis of structural trends in collinear and non-collinear sites or structures. This introduces a new pathway for predicting and systematically analyzing non-collinear magnetic structures, opening opportunities for data-driven exploration of complex magnetic phenomena. This will be covered in future work, with an outlook illustrated in Fig. 7.

Fig. 7 depicts the distribution of coordination numbers and coordination environments for (a) magnetic sites with collinearity $(c) \geq 0.5$ and (b) magnetic sites with $0 \leq c < 0.5$, analyzed with an angle tolerance of $50\,^{\circ}$ in the dataset of unique, commensurate MAGNDATA entries. The two groups display highly different trends in local coordination environments. For example, tetrahedrally coordinated, magnetic sites order almost exclusively with high c. In contrast, the prevalence of eight-fold coordinated sites is 5 times larger in the highly non-collinear dataset compared to the dataset of sites with $c \geq 0.5$. The trends partly coincide with known structural trends in non-collinear magnetism: E.g., REs more often exhibit both the large coordination number of 8[71] and non-collinear magnetism than TMs. Also, some well-known geometrically frustrated structure types, such as pyrochlore, contain 8-fold coordinated sites. Interestingly, most trends remain when these known dependencies are eliminated from the analysis. We believe that structural trends

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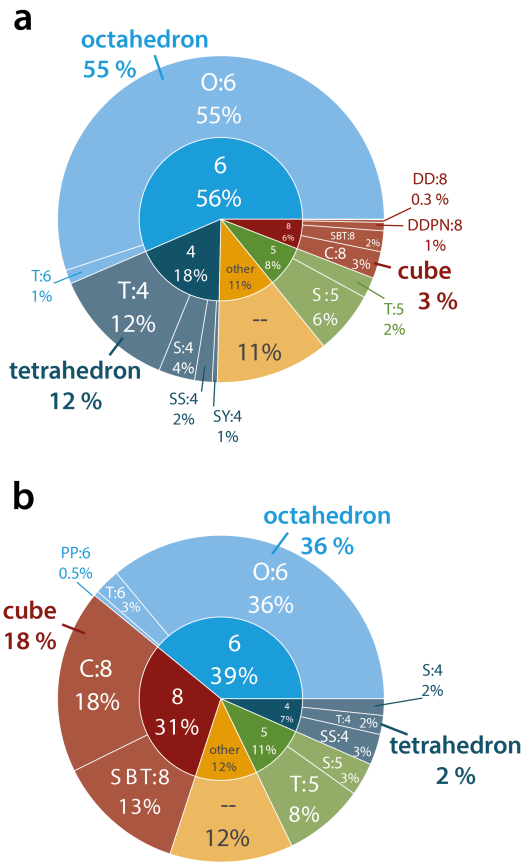

Figure 7: Coordination number – coordination environment sunburst plot of all sites with a collinearity score ≥ 0 (tolerance to count magnetic moment vectors as parallel or antiparallel = 50 °) in the structurally unique, commensurate MAGNDATA dataset. (a) $c \geq 0.5$ and (b) $0 \leq c < 0$. Absolute occurrences (per magnetic primitive cell) are shown. 4375 magnetic sites contribute to (a), 852 contribute to (b). Naming of the coordination environments follows the *ChemEnv* convention:[33] C:8 = cube, DD:8 = dodecahedron with triangular faces, DDPN:8 = dodecahedron with triangular faces - p2345 plane normalized, O:6 = octahedron, PP:6 = pentagonal pyramid, SBT:8 = square-face bicapped trigonal prism, S:4 = square plane, S:5 = square pyramid, SS:4 = see-saw, SY: square non-coplanar, T:4 = tetrahedron, T:5 = trigonal bipyramid, T:6 trigonal prism

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on different length scales must be combined to derive markers for non-collinear magnetism, ranging from local to global metrics.

Our newly introduced site-wise and structure-wise scores provide a simple yet effective means to analyse non-collinearity on different length scales, and the calculation with different angle tolerances enables the analysis of non-collinearity of varying "strengths". In future work, we aim to generalize structural trends of collinear and non-collinear magnetic structures to provide an entry point for the targeted synthesis of new, non-collinear magnetic structures.

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Supporting information

The following files are available free of charge.

- supporting_information.pdf: Additional figures and tables of the statistical analysis and machine learning, methodology details

The Supporting Information contains the following additional citations.[72–81]

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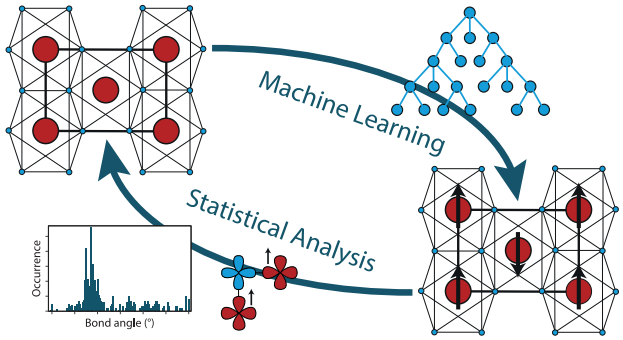

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